

A novel curcumin-based vaginal cream Vacurin selectively eliminates apposed human cervical cancer cells.



OBJECTIVE: Human papillomavirus (HPV) infections remain a leading cause of mortality worldwide. In the U.S. strategies via screening and vaccination prevent HPV-associated cervical neoplasms, but consume immense healthcare costs. The spice component curcumin has potent anticancer and antiviral properties, which have been difficult to harness as a treatment, due to its poor systemic bioavailability. This project tests the possibility of developing a curcumin-based therapy for cervical cancer. **METHODS:** Using four HPV(+) cervical cancer cell lines and normal fibroblasts we first tested the selectivity and potency of curcumin in eliminating HPV(+) cells. Subsequently, we developed a curcumin-based cervical cream and tested its efficacy in eliminating apposed HPV(+) cells and also its possible side effects on the vaginal epithelium of healthy mice. **RESULTS:** Curcumin selectively eliminates a variety of HPV(+) cervical cancer cells (HeLa, ME-180, SiHa, and SW756), suppresses the transforming antigen E6, dramatically inhibits the expression of the pro-cancer protein epidermal growth factor receptor (EGFR), and concomitantly induces p53. Additionally, Vacurin, a uniform colloidal solution of curcumin in a clinically used amphipathic vaginal cream, eliminates apposed HeLa cells while suppressing the expression of EGFR. In mice, daily intravaginal application of Vacurin for three weeks produced no change in body weight and when the mice were sacrificed, the vaginal tract epithelium showed no Vacurin-evoked adverse effects. **CONCLUSION:** We have developed a curcumin-based vaginal cream, which effectively eradicates HPV(+) cancer cells and does not affect non-cancerous tissue. Our preclinical data support a novel approach for the treatment of cervical HPV infection. Copyright © 2012 Elsevier Inc. All rights reserved.